

Machine Learning

CSBB 311

Submitted To:

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Experiment 1

**Data loading, cleaning,
preprocessing, and
visualization using Pandas,
NumPy, and
Matplotlib**

Experiment - 1:

Dataset Used

- Dataset Name: Student Performance
- Source Link: <https://archive.ics.uci.edu/ml/datasets/student+performance>

Background

- Data Loading: Reading data from CSV or Excel using Pandas.
- Cleaning: Handling missing/null values, duplicates, and irrelevant columns.
- Preprocessing: Converting categorical to numerical, scaling values, handling outliers.
- Visualization: Use plots (histograms, boxplots, heatmaps, scatter plots) to understand distribution, correlation, and trends.

Methodology

- a. Load the dataset using `pandas.read_csv()`.
- b. Explore the data using `.info()`, `.describe()`, `.head()`.
- c. Identify and handle missing values.
- d. Convert categorical variables into numerical using `pd.get_dummies()`.
- e. Normalize or scale data if required.
- f. Visualize distributions and relationships using `matplotlib.pyplot` and `seaborn`

Code:

Importing libraries:

```
import pandas as pd
from sklearn.preprocessing import StandardScaler
import matplotlib.pyplot as plt
import seaborn as sns
```

a.

```
df = pd.read_csv("student-mat.csv", sep=";")
```

b.

```
print("First 5 rows:")
print(df.head())
print("\nDataset Information:")
df.info()
print("\nStatistical Summary:")
print(df.describe())
```

c.

```
print(df.isnull().sum().sum()) #this returns 0 which means there are no missing values
```

d.

```
print("Categorical columns:")
print(df.select_dtypes(include="object").columns)
df = pd.get_dummies(df, drop_first=True)
print("\nDataset after encoding:")
print(df.head())
```

e.

```
scaler = StandardScaler()
df_scaled = pd.DataFrame(
    scaler.fit_transform(df),
    columns=df.columns
)
print(df_scaled.head())
```

f.

```
plt.figure(figsize=(8, 5))
sns.histplot(df["G3"], bins=20, kde=True)
plt.title("Distribution of Final Grades")
plt.xlabel("Final Grade (G3)")
plt.ylabel("Number of Students")
plt.show()
```

Output:

b.

```
First 5 rows:
  school sex  age address famsize Pstatus  Medu  Fedu  Mjob  Fjob  ...  \  20 higher  395 non-null  object
0  GP  F  18      U  GT3      A      4      4  at_home teacher ...  21 internet  395 non-null  object
1  GP  F  17      U  GT3      T      1      1  at_home  other ...  22 romantic  395 non-null  object
2  GP  F  15      U  LE3      T      1      1  at_home  other ...  23 famrel   395 non-null  int64
3  GP  F  15      U  GT3      T      4      2  health  services ...  24 freetime  395 non-null  int64
4  GP  F  16      U  GT3      T      3      3  other    other ...  25 goout    395 non-null  int64
                                     26 Dalc     395 non-null  int64
                                     27 Walc     395 non-null  int64
                                     28 health   395 non-null  int64
                                     29 absences  395 non-null  int64
                                     30 G1       395 non-null  int64
                                     31 G2       395 non-null  int64
                                     32 G3       395 non-null  int64
dtypes: int64(16), object(17)
memory usage: 102.0+ KB

[5 rows x 33 columns]

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 395 entries, 0 to 394
Data columns (total 33 columns):
#   Column      Non-Null Count  Dtype
---  -
0  school      395 non-null    object
1  sex          395 non-null    object
2  age          395 non-null    int64
3  address      395 non-null    object
4  famsize      395 non-null    object
5  Pstatus      395 non-null    object
6  Medu         395 non-null    int64
7  Fedu         395 non-null    int64
8  Mjob         395 non-null    object
9  Fjob         395 non-null    object
10 reason      395 non-null    object
11 guardian    395 non-null    object
12 traveltime  395 non-null    int64
13 studytime   395 non-null    int64
14 failures    395 non-null    int64
15 schoolsup   395 non-null    object
16 famsup      395 non-null    object
17 paid        395 non-null    object
18 activities  395 non-null    object
19 nursery     395 non-null    object

Statistical Summary:
      age      Medu      Fedu  traveltime  studytime  failures \
count  395.000000  395.000000  395.000000  395.000000  395.000000  395.000000
mean    16.696203    2.749367    2.521519    1.448101    2.035443    0.334177
std     1.276043    1.094735    1.088201    0.697505    0.839240    0.743651
min     15.000000    0.000000    0.000000    1.000000    1.000000    0.000000
25%    16.000000    2.000000    2.000000    1.000000    2.000000    0.000000
50%    17.000000    3.000000    2.000000    1.000000    2.000000    0.000000
75%    18.000000    4.000000    3.000000    2.000000    2.000000    0.000000
max     22.000000    4.000000    4.000000    4.000000    4.000000    3.000000

      famrel  freetime  goout  Dalc  Walc  health \
count  395.000000  395.000000  395.000000  395.000000  395.000000  395.000000
mean    3.944304    3.235443    3.108861    1.481013    2.291139    3.554430
std     0.896659    0.998862    1.113278    0.890741    1.287897    1.390383
min     1.000000    1.000000    1.000000    1.000000    1.000000    1.000000
25%     4.000000    3.000000    2.000000    1.000000    1.000000    3.000000
50%     4.000000    3.000000    3.000000    1.000000    2.000000    4.000000
75%     5.000000    4.000000    4.000000    2.000000    3.000000    5.000000
max     5.000000    5.000000    5.000000    5.000000    5.000000    5.000000

      absences  G1  G2  G3
count  395.000000  395.000000  395.000000  395.000000
mean    5.708861    10.908861    10.713924    10.415190
std     8.003096    3.319195    3.761505    4.581443
min     0.000000    3.000000    0.000000    0.000000
25%     0.000000    8.000000    9.000000    8.000000
50%     4.000000    11.000000    11.000000    11.000000
75%     8.000000    13.000000    13.000000    14.000000
max    75.000000    19.000000    19.000000    20.000000
```

c.

```
np.int64(0)
```

d.

Categorical columns:
Index(['school', 'sex', 'address', 'famsize', 'Pstatus', 'Mjob', 'Fjob',
'reason', 'guardian', 'schoolsup', 'famsup', 'paid', 'activities',
'nursery', 'higher', 'internet', 'romantic'],
dtype='object')

Dataset after encoding:

	age	Medu	Fedu	traveltime	studytime	failures	famrel	freetime	goout
0	18	4	4	2	2	0	4	3	4
1	17	1	1	1	2	0	5	3	3
2	15	1	1	1	2	3	4	3	2
3	15	4	2	1	3	0	3	2	2
4	16	3	3	1	2	0	4	3	2

	Dalc	...	guardian_mother	guardian_other	schoolsup_yes	famsup_yes
0	1	...	True	False	True	False
1	1	...	False	False	False	True
2	2	...	True	False	True	False
3	1	...	True	False	False	True
4	1	...	False	False	False	True

	paid_yes	activities_yes	nursery_yes	higher_yes	internet_yes
0	False	False	True	True	False
1	False	False	False	True	True
2	True	False	True	True	True
3	True	True	True	True	True
4	True	False	True	True	False

	romantic_yes
0	False
1	False
2	False
3	True
4	False

[5 rows x 42 columns]

e.

	age	Medu	Fedu	traveltime	studytime	failures	famrel
0	1.023046	1.143856	1.360371	0.792251	-0.042286	-0.449944	0.062194
1	0.238380	-1.600009	-1.399970	-0.643249	-0.042286	-0.449944	1.178860
2	-1.330954	-1.600009	-1.399970	-0.643249	-0.042286	3.589323	0.062194
3	-1.330954	1.143856	-0.479857	-0.643249	1.150779	-0.449944	-1.054472
4	-0.546287	0.229234	0.440257	-0.643249	-0.042286	-0.449944	0.062194

	freetime	goout	Dalc	...	guardian_mother	guardian_other
0	-0.236010	0.801479	-0.540699	...	0.668496	-0.296908
1	-0.236010	-0.097908	-0.540699	...	-1.495896	-0.296908
2	-0.236010	-0.997295	0.583385	...	0.668496	-0.296908
3	-1.238419	-0.997295	-0.540699	...	0.668496	-0.296908
4	-0.236010	-0.997295	-0.540699	...	-1.495896	-0.296908

	schoolsup_yes	famsup_yes	paid_yes	activities_yes	nursery_yes
0	2.597133	-1.257656	-0.919671	-1.017881	0.507899
1	-0.385040	0.795130	-0.919671	-1.017881	-1.968894
2	2.597133	-1.257656	1.087346	-1.017881	0.507899
3	-0.385040	0.795130	1.087346	0.982433	0.507899
4	-0.385040	0.795130	1.087346	-1.017881	0.507899

	higher_yes	internet_yes	romantic_yes
0	0.23094	-2.232677	-0.708450
1	0.23094	0.447893	-0.708450
2	0.23094	0.447893	-0.708450
3	0.23094	0.447893	1.411533
4	0.23094	-2.232677	-0.708450

[5 rows x 42 columns]

f.

